

Community detection (CD) - project assignment

Environment for enhanced performance evaluation of the selected community detection algorithms, which should include:

GRAPHS:

- The comparison of at least 4 algorithms for graphs,
- The comparison provided for at least 3 real-world and 2 synthetic datasets (graphs with different characteristics),
- The usage of different performance measures:
 - modularity function for graphs (e.g., from [igraph](#))
 - AMI, ARI (when ground-truth communities are known) (e.g., from [sklearn](#)),
- Bayesian optimization
(e.g. based on <https://github.com/bayesian-optimization/BayesianOptimization>)
for selected hyperparameters of one of the algorithms
(e.g., *resolution* parameter of an igraph implementation of the Leiden algorithm:
https://python.igraph.org/en/stable/api/igraph.GraphBase.html#community_leiden).

HYPERGRAPHS:

- The comparison provided for at least 2 real-world and 2 synthetic datasets,
- Each of the hypergraphs should be transformed into a 2-section graph,
- Then, a similar comparison as for graphs should be provided (for at least 4 algorithms),
- The usage of different performance measures:
 - modularity function for hypergraphs (e.g. strict modularity from [hypernetx](#)),
 - AMI, ARI.

The materials, links, and solutions for the basic assignment may be useful when building the project results.